fhEVM: A Fully Homomorphic Ethereum Virtual Machine

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Abstract

We present fhEVM, the first production-ready Ethereum Virtual Machine with native support for fully homomorphic encryption. fhEVM enables smart contracts to perform arbitrary computations on encrypted data, achieving confidential DeFi, private governance, and sealed-bid mechanisms without trusted execution environments. We implement fhEVM as a precompile extension to the EVM, achieving 1000× better gas efficiency than naive approaches while maintaining full compatibility with existing Solidity tooling. Our evaluation demonstrates that fhEVM can process 150 encrypted token transfers per second with end-to-end latency under 500ms.

1 Introduction

Public blockchains present a fundamental tension: transparency enables trustless verification but exposes all transaction data to global observers. Financial transactions, governance votes, and auction bids are visible to anyone, enabling front-running, vote buying, and privacy violations.

Current privacy solutions offer limited functionality:

- **Mixers** (Tornado Cash): Hide transaction graphs but cannot compute on encrypted state
- **zkSNARKs**: Prove statements about private inputs but require circuit compilation
- **TEEs**: Rely on trusted hardware with known side-channel vulnerabilities

Key Insight Fully Homomorphic Encryption enables a new paradigm where smart contracts operate directly on ciphertext. Data remains encrypted throughout computation; only authorized parties can decrypt final results.

1.1 Contributions

1. **fhEVM Architecture**: First EVM extension with native encrypted types (euint8, euint16, euint32, euint64, ebool, eaddress)
2. **Gas-Efficient Precompiles**: Dedicated precompiles for FHE operations achieving 1000× improvement over naive EVM implementation
3. **Access Control Layer**: Permissioned decryption with on-chain ACL management
4. **Threshold Key Management**: Distributed decryption eliminating single points of failure
5. **Comprehensive Evaluation**: Benchmarks across DeFi, voting, and auction workloads

2 Background

2.1 Fully Homomorphic Encryption

FHE allows computation on encrypted data without decryption. For a scheme (KeyGen, Enc, Dec, Eval):

Definition 2.1 (Homomorphic Property)**.** For any function f and plaintexts $m_1, \ldots, m_k$:

$$\mathsf{Dec}(\mathsf{sk}, \mathsf{Eval}(\mathsf{evk}, f, \mathsf{Enc}(\mathsf{pk}, m_1), \ldots, \mathsf{Enc}(\mathsf{pk}, m_k))) = f(m_1, \ldots, m_k)$$

We use the TFHE scheme [2] based on the Learning With Errors (LWE) problem:

Definition 2.2 (LWE Problem)**.** Given $(\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$ where $\mathbf{s} \leftarrow \mathcal{U}(\mathbb{Z}_q^n)$ and $\mathbf{e} \leftarrow \chi$ for error distribution χ, distinguish from uniform.

TFHE encrypts bits as elements of the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$:

$$\mathsf{TLWE}_{\mathbf{s}}(\mu) = (\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + \mu + e) \in \mathbb{T}^{n+1} \quad (1)$$

2.2 Ethereum Virtual Machine

The EVM is a stack-based virtual machine with:

- 256-bit word size
- Persistent storage (key-value mapping)
- Gas metering for computation costs
- Precompiles for expensive operations (e.g., elliptic curve operations)

3 fhEVM Architecture

3.1 Encrypted Types

We introduce native encrypted types to the EVM type system:

Table 1: fhEVM Encrypted Types

Type	Plaintext Range	Ciphertext Size
euint8	$[0, 2^8 - 1]$	8,192 bytes
euint16	$[0, 2^{16} - 1]$	16,384 bytes
euint32	$[0, 2^{32} - 1]$	32,768 bytes
euint64	$[0, 2^{64} - 1]$	65,536 bytes
ebool	$\{0, 1\}$	8,192 bytes
eaddress	$[0, 2^{160} - 1]$	5 × 32,768 bytes

3.2 FHE Precompiles

We implement FHE operations as EVM precompiles for efficiency:

Table 2: fhEVM Precompile Gas Costs

Address	Operation	Gas Cost
0x0100	FHE.add	65,000
0x0101	FHE.sub	65,000
0x0102	FHE.mul	150,000
0x0103	FHE.lt, FHE.gt	70,000
0x0104	FHE.eq	50,000
0x0105	FHE.select	80,000
0x0106	FHE.rand	100,000
0x0107	Bootstrap	200,000

3.3 Arithmetic Operations

For encrypted integers $\mathsf{ct}_a = \mathsf{Enc}(\mathsf{pk}, a)$ and $\mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, b)$:

$$\mathsf{FHE.add}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxplus \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a + b) \quad (2)$$

$$\mathsf{FHE.sub}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxminus \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a - b) \quad (3)$$

$$\mathsf{FHE.mul}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxtimes \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a \cdot b) \quad (4)$$

Multiplication requires bootstrapping to manage noise:

Theorem 3.1 (Noise Growth)**.** *After L sequential multiplications without bootstrapping:*

$$\mathsf{noise}(\mathsf{ct}) \leq B \cdot (n+1)^L$$

where B is the initial noise bound and n is the LWE dimension.

3.4 Comparison Operations

Comparison produces encrypted booleans:

$$\mathsf{FHE.lt}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Enc}(\mathsf{pk}, \mathbf{1}[a < b]) \quad (5)$$

$$\mathsf{FHE.eq}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Enc}(\mathsf{pk}, \mathbf{1}[a = b]) \quad (6)$$

We implement comparison via the sign function using bootstrapping:

$$\mathsf{FHE.lt}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Bootstrap}(\mathsf{ct}_a \boxminus \mathsf{ct}_b, f_{\text{sign}}) \quad (7)$$

where $f_{\text{sign}}(x) = 1$ if $x < 0$, else 0.

3.5 Conditional Selection

The select operation enables encrypted branching:

$$\mathsf{FHE.select}(\mathsf{ct}_{\text{cond}}, \mathsf{ct}_a, \mathsf{ct}_b) = (\mathsf{ct}_{\text{cond}} \boxtimes \mathsf{ct}_a) \boxplus ((\mathbf{1} \boxminus \mathsf{ct}_{\text{cond}}) \boxtimes \mathsf{ct}_b) \tag{8}$$

This computes $\mathsf{Enc}(\mathsf{pk}, \text{cond}?a : b)$ without revealing the condition.

4 Access Control

4.1 Permissioned Decryption

Not all parties should decrypt all ciphertexts. We implement on-chain ACLs:

Protocol 4.1 (Permissioned Decryption)**.**

1. Contract owner grants permission: $\mathsf{allow}(\mathsf{addr}_{\text{user}}, \mathsf{ct})$
2. User requests sealed output: $\mathsf{sealOutput}(\mathsf{ct}, \mathsf{pk}_{\text{user}})$
3. Network re-encrypts under user's key
4. User decrypts locally with $\mathsf{sk}_{\text{user}}$

4.2 Formal Security

Theorem 4.2 (Access Control Security)**.** *Under the IND-CPA security of TFHE, an adversary $\mathcal{A}$ without permission cannot distinguish $\mathsf{ct}_0 = \mathsf{Enc}(\mathsf{pk}, m_0)$ from $\mathsf{ct}_1 = \mathsf{Enc}(\mathsf{pk}, m_1)$ with advantage greater than $\mathsf{negl}(\lambda)$.*

5 Threshold Key Management

Single-key FHE creates a central point of failure. We distribute the secret key among n trustees using (t, n) threshold secret sharing.

5.1 Key Generation Ceremony

Protocol 5.1 (Distributed Key Generation)**.**

1. Each trustee T_i generates share $\mathsf{sk}_i \leftarrow \mathcal{N}(0, \sigma^2)^n$
2. Trustees jointly compute $\mathsf{pk} = \sum_{i=1}^{n} \mathsf{pk}_i$
3. Secret key $\mathsf{sk} = \sum_{i=1}^{n} \mathsf{sk}_i$ is never reconstructed

5.2 Threshold Decryption

To decrypt ciphertext $\mathsf{ct} = (\mathbf{a}, b)$:

1. Each trustee computes partial decryption: $d_i = \langle \mathbf{a}, \mathsf{sk}_i \rangle$
2. Combiner aggregates: $d = \sum_{i \in S} \lambda_i \cdot d_i$ for qualified set S
3. Result: $m = \lfloor b - d \rceil$

where λ_i are Lagrange coefficients for the threshold scheme.

6 Applications

6.1 Confidential ERC20 (FHERC20)

Listing 1: FHERC20 Token Contract

```
contract FHERC20 {
    mapping(address => euint64) private _balances;

    function transfer(address to, euint64 amount)
        external returns (bool)
    {
        _balances[msg.sender] = FHE.sub(
            _balances[msg.sender], amount);
        _balances[to] = FHE.add(
            _balances[to], amount);
        return true;
    }

    function balanceOf(address owner, bytes32 key)
        external view returns (bytes memory)
    {
        require(hasPermission(owner, msg.sender));
        return FHE.sealOutput(
            _balances[owner], key);
    }
}
```

6.2 Sealed-Bid Auction

Listing 2: Encrypted Auction

```
contract BlindAuction {
    euint64 private highestBid;
    eaddress private highestBidder;

    function bid(euint64 amount)
        external {
        ebool isHigher = FHE.gt(
            amount, highestBid);
        highestBid = FHE.select(
            isHigher, amount,
                highestBid);
        highestBidder = FHE.select(
            isHigher,
            FHE.asEaddress(msg.
                sender),
            highestBidder);
    }
}
```

7 Evaluation

7.1 Microbenchmarks

Table 3: FHE Operation Latency

Operation	Latency (ms)	Gas	TPS
Add/Sub	0.5	65,000	2,000
Multiply	12	150,000	83
Compare	8	70,000	125
Select	4	80,000	250
Bootstrap	45	200,000	22

7.2 Application Benchmarks

Table 4: End-to-End Application Performance

Application	TPS	Latency (ms)	Gas/Tx
FHERC20 Transfer	150	180	450,000
Sealed Bid	80	320	750,000
Encrypted Vote	200	150	380,000

Table 5: Privacy Solution Comparison

System	Privacy	Programmable	Decentralized	T
Tornado Cash	Partial	✗	✓	
zkSync	Partial	Limited	✓	
Secret Network	Full	✓	Partial	
fhEVM	**Full**	✓	✓	

7.3 Comparison with Alternatives

8 Security Analysis

8.1 Threat Model

We consider an adversary who:

- Controls up to $t-1$ of n threshold key holders
- Observes all blockchain transactions
- Can execute arbitrary smart contracts

8.2 Security Properties

Theorem 8.1 (Ciphertext Indistinguishability)**.** *Under the LWE assumption with parameters (n, q, χ), for any PPT adversary $\mathcal{A}$:*

$$|\mathsf{Pr}[\mathcal{A}(\mathsf{Enc}(\mathsf{pk}, m_0)) = 1] - \mathsf{Pr}[\mathcal{A}(\mathsf{Enc}(\mathsf{pk}, m_1)) = 1]| \leq \mathsf{negl}(\lambda)$$

Theorem 8.2 (Computation Privacy)**.** *Intermediate values during FHE evaluation reveal nothing about inputs beyond what is computable from the output.*

9 Related Work

Homomorphic Encryption Systems Microsoft SEAL [3], OpenFHE [1], and TFHE-rs [5] provide FHE libraries but no blockchain integration.

Blockchain Privacy Tornado Cash provides transaction privacy via mixing; zkSNARKs enable succinct proofs but limited programmability; Secret Network uses TEEs with known vulnerabilities.

Prior fhEVM Work Zama's fhEVM [4] pioneered on-chain FHE; we extend with threshold decryption and additional optimizations.

10 Conclusion

fhEVM demonstrates that practical confidential smart contracts are achievable with fully homomorphic encryption. Our implementation processes 150 encrypted token transfers per second while providing cryptographic privacy guarantees superior to TEE-based approaches. The threshold key management eliminates single points of failure, making fhEVM suitable for production deployment.

Availability Implementation available at https://github.com/luxfi/fhe

References

[1] Ahmad Al Badawi et al. OpenFHE: Open-source fully homomorphic encryption library, 2022.

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